Linear Search:

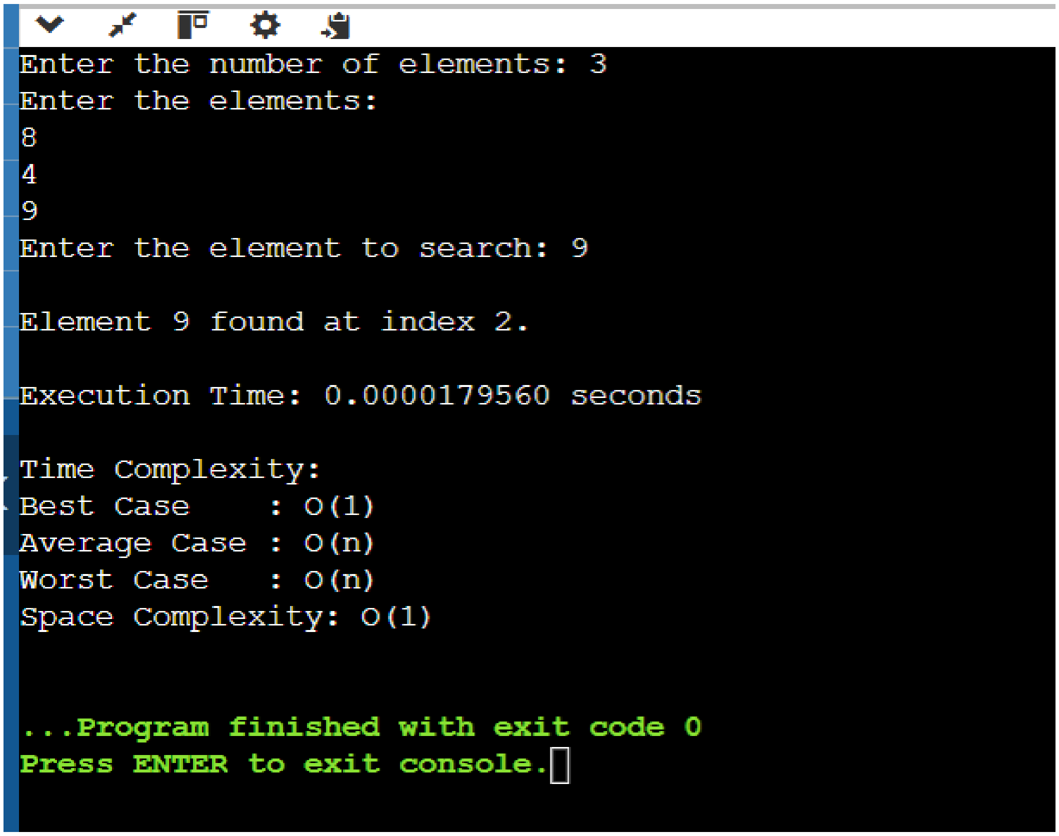
Linear Search is the simplest searching algorithm. It checks each element of the array or list one by one until the target element is found or the end of the array is reached. It works on both sorted and unsorted data.

Algorithm:

```
LinearSearch(arr, target)

1. Start
2. For i = 0 to length(arr) - 1
      a. If arr[i] == target
            Return i
3. Return -1 (Element not found)
4. Stop
```

Output:



Time Complexity:

- **Best Case: O(1)**
  The target element is found at the first position.
- **Average Case: O(n)**
  The target element is found somewhere in the middle of the array.
- **Worst Case: O(n)**
  The target element is at the last position or is not present in the array.

Space Complexity:

- **Iterative Linear Search: O(1)**
  Uses only a few variables for searching.
- **Recursive Linear Search: O(n)**
  Uses additional memory due to the recursive call stack.